Trading rulebook

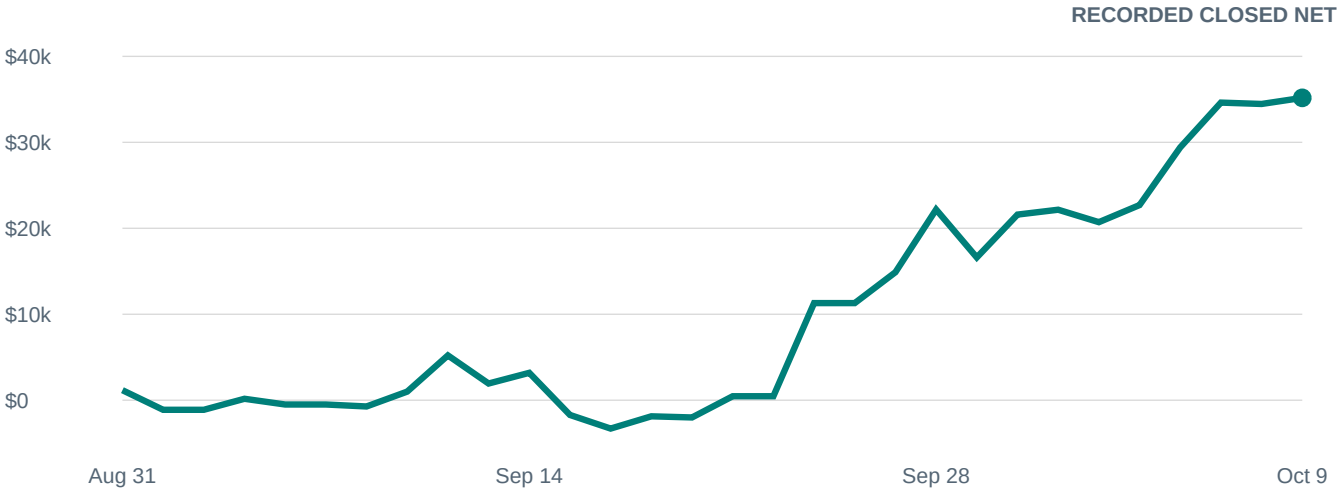
ESB_q

ES futures reclaim strategy
Rules, sizing, position management and forward results

Revised policy 10 October 2026 | Historical results: OLD policy, 31 Aug - 9 Oct

ESB_q identifies a confirmed swing leg, watches for price to reclaim the start of that leg, and trades the reclaim when the previous bar agrees with an EMA50 trend filter. It holds the full position behind a stepped trailing stop, with a full exit at the 2.5U target if reached first. This guide explains the **current implemented book**, including its accounting conventions and important limitations.

OLD policy trades	OLD policy win rate	OLD policy net
37	59.5%	$35,172.50



The curve is cumulative closed-trade shadow P&L, after the book's modeled $4 round-trip commission per contract. It is not a broker account equity curve.

The essentials

15-minute structure. Five bars on either side confirm a swing; the leg must be at least one ATR14 and five ES points. A long reclaims a down leg; a short reclaims an up leg.

One position at a time. A prior completed-bar excursion arms a 0.25U step / 0.25U lag ratchet. All contracts exit together. There are no partial exits and no separately managed runner.

New admission cap. Final entry-to-stop exposure, modeled stop slippage and fees must fit within $5,000 and 10 contracts. No affordable contract means no entry. The 2.5U target now exits the full position. Realized gap losses can still exceed planned risk.

Version boundary: every result above belongs to the original policy. The revised policy is code-tested and loaded after hours, with no forward trades yet.

How the book finds a trade



Illustration only. U is the size of the confirmed swing leg; it is not a statistical standard deviation or a GEX distance.

1 Build the price history

The current process samples ES midpoint observations approximately every 15 seconds into 15-minute OHLC buckets. It calculates over the latest 400 stored bars, including retained outside-RTH bars, and requires at least 60 bars before entry evaluation. The entry window is RTH, but indicator history is not RTH-only.

2 Confirm the extremes

A pivot high must be strictly higher than the five highs before it and the five highs after it. A pivot low must be strictly lower than all ten surrounding lows. Equal highs or lows do not qualify. Five later completed bars are needed, so confirmation normally arrives 75 minutes after the pivot bar closes.

3 Select the most recent qualifying leg

ATR14 measures typical bar movement, including gaps, using Wilder smoothing. EMA50 weights recent closes more heavily and provides the trend reference.

Pair each new confirmed pivot with the latest earlier opposite pivot. The earlier price is **zero**; the new extreme is **one**; **U = abs(one - zero)**. A qualifying leg has positive U and U at least equal to Wilder ATR14 at the confirmation bar. The newest qualifying leg replaces the prior one; an unqualified new leg does not erase an older valid leg.

Leg	Trade direction	Reclaim watched
one below zero	LONG	Bar high reaches or exceeds zero
one above zero	SHORT	Bar low reaches or falls below zero

The structural stop remains at one. The book assigns no separate age expiry to a leg. It does not use GEX, sigma bands, volume, VWAP, SPY/QQQ agreement or a news filter to choose a trade.

The exact entry checklist

Capacity and identity

The book must be flat, have at least 60 bars, and have a qualifying leg. The signal ID combines ESB_q with the extreme pivot timestamp. Any existing record with that ID prevents another attempt.

Price trigger

For a long, the completed bar high must reach zero; for a short, its low must reach zero. A closing-price reclaim is not required, and the test does not require a newly observed crossing from the other side.

Session and calendar

The trigger bar START must be at or after 09:30 and before 16:00 New York time. Holidays and early-close dates in the current calendar block new entries. Weekend sampling is blocked.

Minimum leg size

U must be at least 5.0 points. A smaller triggered leg is permanently recorded as skip_smallstop.

Previous bar trend

Use the close and EMA50 of the bar immediately BEFORE the trigger bar. A long needs close > EMA50. A short needs close <= EMA50. A failure is permanently recorded as skip_counter_trend.

Model entry and size

If zero lies inside the trigger bar range, model entry is zero. Otherwise use the bar open. Compute desired size from model geometry, then price the entry and enforce the final cost-inclusive risk and contract caps. Reject an entry at or beyond the target.

What happens after a rejection

Small-stop, counter-trend, risk-budget and passed-target rejections consume that leg ID; it will not be reconsidered even if conditions later improve. No-trigger, out-of-session and holiday checks do not consume the leg. While a position is open, new trades are blocked.

At each bar close, management runs before entry evaluation. A different leg can therefore enter in the same cycle after an old trade exits. A new trade starts with MFE and MAE at zero; it is not managed again against its own trigger bar in that cycle.

How much the book puts at risk

ESB_q trades whole futures contracts. It does not invest a fixed cash premium. Broker margin is not stored in this ledger, and contract notional is not the cash needed to hold the position.

1 Calculate desired quantity

```
base = max(1, min(10, floor(5000 / (U * 50))))
```

The existing 1.5x modest-gap multiplier can increase desired size when the model entry is within 0.75U beyond zero. Desired size is only a request; it cannot override the final risk or quantity limits.

2 Price the entry, then enforce the caps

```
distance = direction * (booked_entry - original_stop)
unit_risk = (distance + 0.08 + max(0.25, stop_slip)) * 50
quantity = min(desired, 10, floor(5000 / unit_risk))
planned_risk = quantity * unit_risk
```

Direction is +1 for long and -1 for short. With current defaults, costs add 0.33 points ($16.50) per contract to the entry-to-stop distance. Reapply both limits after the boost and quote-adjusted entry. Do not move the structural stop to fit more contracts.

If quantity is zero, skip the trade. Non-finite risk inputs, invalid stop geometry and a booked entry at or beyond the fixed target are rejected. Accepted quantity is an integer from 1 to 10; planned risk is at most $5,000.

Examples using zero 7800, stop 7780 and U 20

Booked entry	Desired / admitted	Final planned risk
7800, clean	5 / 4	$4,066.00
7810, modest gap	7 / 3	$4,549.50
One-contract risk over $5,000	Any / 0	Skip; no forced minimum

What the cap means during a trade

Quantity is fixed after entry; the book never increases it while the trade is open. The structural stop and ratchet remain active, and all contracts exit together. A market gap, delayed evaluation or stale-data fallback can produce a loss larger than the entry-time plan; $5,000 is not a guaranteed realized-loss ceiling.

Before the repair, the 37 recorded trades used 1-12 contracts (average 5.49). Their average recorded U-risk was $4,527.27, while average fill-to-original-stop price exposure was $8,605.31 before costs. Those historical values are preserved, not rewritten.

How the open trade is managed

Zero, one, U and quantity are frozen at entry. A later detected swing does not replace the position's anchors. The book retains the full quantity until a ratchet, structural-stop, 2.5U target or time exit. New entries store their policy and cost settings; existing legacy rows keep their original ratchet-only policy.

The stepped ratchet

Favorable excursion is measured from **zero**, not from the booked entry. On a long it is bar high minus zero; on a short it is zero minus bar low. The active stop uses the best excursion stored BEFORE the current bar.

```
peakU = prior_MFE / U
n = int(peakU / 0.25)
ratchet = zero + direction * (n * 0.25 - 0.25) * U
```

The ratchet is armed only when peakU is at least 0.25. Direction is +1 for long and -1 for short. The current bar's new excursion is saved for the next bar; it cannot raise a stop retrospectively within the same bar.

Prior best excursion	Long stop when zero 7800 and U 20
Below 0.25U	Original structural stop at 7780
0.25U to below 0.50U	7800
0.50U to below 0.75U	7805
0.75U to below 1.00U	7810
1.00U to below 1.25U	7815
Target at zero + 2.50U	7850: full exit; protection has priority on ambiguous bars

Exit priority for each completed bar

First: if an armed ratchet is crossed, exit all contracts. **Second:** otherwise exit all if the original one-level stop is crossed. **Third:** otherwise exit all when the bar touches the 2.5U target. **Fourth:** otherwise apply the calendar time exit when the bar start qualifies. If none applies, hold and update excursion records.

A ratchet at zero is not necessarily break-even after entry gaps and costs. There is no partial take-profit, fixed holding limit, EMA-flip exit or daily loss breaker. When both protection and target occur in the same bar, protection wins; the model does not assume the favorable event happened first. MFE and MAE use whole-bar extremes and are not exact tick-path measurements after an intra-bar exit.

Monitoring and exit valuation

Between bar closes

Approximately every 15 seconds, the process records the first sampled quote at a relevant entry, stop, ratchet or target crossing. Quotes are tied to the current bar bucket; entry quotes are also tied to the leg. The strategy commits its entry and exit records on completed-bar evaluation, rather than submitting trades at each sampled crossing.

How the shadow fill is priced

Situation	Valuation rule
Fresh entry quote	Long pays max(model, ask); short receives min(model, bid).
Fresh stop or ratchet	Use the adverse capped quote. If the bar opened strictly beyond the stop against the position, use the raw exit-side quote.
Fresh target exit	Long receives min(target, bid); short pays max(target, ask). No optimistic price improvement is assumed.
Fresh time exit	Use the raw exit-side bid for a long sale or ask for a short cover.
Stale or absent quote	Use the worst of model plus an adverse tick and eligible available quote/last candidates; label stale_conservative.
Fresh but implausible quote	If the side price is more than 2% from model, use the conservative path and label quote_insane.

Fresh means both quote sides are present, ask is not below bid, and the recorded source age is at most five seconds. The age is not a verified exchange-event latency. A captured snapshot retains its capture-age value. All 37 current trades carry clean quote-quality flags on both entry and exit.

Session flattening and the clock fallback

Normal flat reference is 16:00 ET. The bar-driven check uses the bar's START time, so the 15:45 bar does not itself meet that cutoff when it completes. A separate clock failsafe is eligible at 16:02, or when an open position belongs to a prior date. It settles from the last completed close plus the quote-valuation policy. Calendar-blocked days use a 12:45 reference and 12:47 failsafe.

A snapshot near 15:59 is only usable when its bar bucket matches the settlement bucket. It is not a guaranteed 16:00 fill. The actual result depends on process availability and polling; stale fallback remains modeled valuation.

Net profit calculation

```
net_points = direction * (exit - entry) - 0.08
net_dollars = net_points * $50 * contracts
```

Commission is $4 per contract round trip. Fresh-quote pricing adds no extra stop-slip tick; the spread is already represented by bid/ask. The original book does not enforce displayed depth, simulate partial fills or round every model-capped fill to the tick.

Parameter reference

Revised prospective policy: esbq_risk_target_20261010. Loaded risk cap $5,000; max contracts 10. Historical results used the prior policy.

Parameter	Value	Meaning
Bar interval	900 seconds	15-minute sampled-midpoint bars
Calculation history	400 bars	Includes retained outside-RTH history
Minimum history	60 bars	EMA50 + pivot5 + 5
Pivot strength	5 each side	Strict inequality; five later bars confirm
ATR period and minimum	14; 1.0x	Wilder ATR at pivot confirmation
Minimum U	5 points	Triggered smaller leg is permanently skipped
EMA period	50	Previous bar close compared with previous EMA
Entry session	09:30 to <16:00	New York time; bar-start gate
Entry mode	limit	Touch/reclaim geometry; no closing reclaim required
Point value	$50 per point	ES accounting multiplier
Planned-risk cap	$5,000	Final entry-to-stop distance plus modeled costs
Final contracts	1 to 10	Reapplied after boost; zero affordable means skip
Gap size boost	1.5x	Only non-clean model gaps within 0.75U
Position capacity	1	No pyramiding; no daily trade count cap
Exit mode	ratchet + target	Entire quantity exits together
Step and lag	0.25U / 0.25U	Based on prior stored zero-anchored MFE
Active target	2.5U	Full exit after protective checks
Commission	0.08 points	Equals $4 round trip per contract
Fallback adjustment	0.25 points	One adverse tick under default settings
Quote freshness	5 seconds	Recorded quote-source age
Quote sanity band	2%	Side price versus model price
Price-input freshness	60 seconds	Separate bar-sampling acceptance gate
Poll interval	15 seconds	Approximate; processing adds elapsed time
Normal flat reference	16:00 ET	Clock failsafe eligible two minutes later
Position update throttle	1,200 seconds	Checked at bar closes; reporting only
Heartbeat	7,200 seconds	Reporting only
Daily recap	From 16:05 ET	Once per weekday; reporting only

Historical forward results

OLD policy only. Recorded closed outcomes by the ledger's session date. Current epoch only; retired books are excluded. Blank-activity dates mean no recorded trade, not independently verified all-day uptime.

2026 date	Trades	W / L	Daily net	Cumulative net
08-31	1	1 / 0	$1,167.50	$1,167.50
09-01	2	1 / 1	-$2,282.00	-$1,114.50
09-02 *	0	0 / 0	$0.00	-$1,114.50
09-03	2	1 / 1	$1,281.00	$166.50
09-04	1	0 / 1	-$665.00	-$498.50
09-07 *	0	0 / 0	$0.00	-$498.50
09-08	3	2 / 1	-$229.50	-$728.00
09-09	1	1 / 0	$1,730.00	$1,002.00
09-10	1	1 / 0	$4,210.00	$5,212.00
09-11	1	0 / 1	-$3,270.00	$1,942.00
09-14	2	1 / 1	$1,244.00	$3,186.00
09-15	2	0 / 2	-$4,899.50	-$1,713.50
09-16	2	1 / 1	-$1,586.00	-$3,299.50
09-17	1	1 / 0	$1,426.50	-$1,873.00
09-18	1	0 / 1	-$133.00	-$2,006.00
09-21	1	1 / 0	$2,468.00	$462.00
09-22 *	0	0 / 0	$0.00	$462.00
09-23	1	1 / 0	$10,835.00	$11,297.00
09-24 *	0	0 / 0	$0.00	$11,297.00
09-25	2	2 / 0	$3,593.00	$14,890.00
09-28	1	1 / 0	$7,292.50	$22,182.50
09-29	1	0 / 1	-$5,574.00	$16,608.50
09-30	3	2 / 1	$4,985.00	$21,593.50
10-01	1	1 / 0	$567.00	$22,160.50
10-02	1	0 / 1	-$1,449.00	$20,711.50
10-05	1	1 / 0	$1,989.00	$22,700.50
10-06	1	1 / 0	$6,710.00	$29,410.50
10-07	1	1 / 0	$5,210.00	$34,620.50
10-08	1	0 / 1	-$158.00	$34,462.50
10-09	2	1 / 1	$710.00	$35,172.50

* September 2, 22 and 24: no recorded trade. September 7: holiday entries blocked. Weekends omitted. No open positions or missing closed P&L were found at the snapshot.

Reading the performance correctly

Forward measure	Recorded value
Period	31 August to 9 October 2026
Closed trades	37: 22 wins, 15 losses, no break-even trades
Win rate	59.46%
Net after modeled commission	$35,172.50
Gross before modeled commission	$35,984.50
Modeled commission	$812.00
Profit factor	2.23
Average net per trade	$950.61
Largest closed-trade drawdown	$8,511.50
Best and worst trade	$10,835.00 / -$5,574.00
Exit mix	29 ratchet; 5 time; 3 structural stop
Trading days	26 active: 16 positive and 10 negative

What these results establish

Every one of the 37 recorded trade outcomes reconciles with its entry price, exit price, direction, quantity and $4 round-trip commission. The daily table reconciles to the cumulative total. The sample contains 24 clean bar-geometry entries and 13 gap entries; both categories can have clean quote-quality flags.

What the results do not establish

These are simulated forward records, not broker statements. The model assumes full quantity, does not certify contract identity or queue liquidity at each fill, and permits conservative modeled settlement when evidence is stale. The drawdown above uses closed trades only, not a continuously marked equity curve. A positive six-week sample is not proof of future performance.

Source and companion specification

Rules were checked against the active ESB_q definition and entry/management logic in **es15_sd_books.py**, the confirmed-leg and indicator helpers in **sd_shadow.py**, and **market_calendar.py**. Results were read from a consistent, read-only snapshot of the ESB_q epoch ledger on 10 October 2026. The October 10 repair was installed and the ESB_q worker reloaded. Historical ledger rows were verified unchanged; none of these results are a backtest or forward result of the repair.

The companion LLM specification provides exact formulas, decision order, calendar dates, quote-accounting cases and reconstruction tests. It contains no API configuration or broker order-routing instructions.

Repair audit and operating limits

Change	Verified behavior
Final risk sizing	Whole quantity recomputed from booked entry to original stop, plus fees and stop allowance. Boost cannot bypass either cap.
Active target	Long and short 2.5U exits tested; prior ratchet and structural stop win ambiguous same-bar conflicts.
Policy persistence	New trades snapshot policy/costs. Legacy rows remain ratchet-only. Reopened test database retains the saved policy.
Position messages	Position updates show the active protective level and whether the target is active or legacy reference-only.
Deployment	Verified source identity loaded by one replacement worker; old trade rows, bars and quote/forming state preserved.

What was tested

Fourteen focused regression tests cover final quote-adjusted sizing, boosted gaps, unaffordable entries, risk bounds in both directions, invalid inputs, target selection, ambiguous bars, prior-bar ratchets, legacy policies, restart persistence, quote snapshots and additive schema migration. The existing nine-part quote-accounting suite also passes.

What remains to be observed

The revised policy has no natural forward trades yet. It may reduce profit as well as risk by admitting fewer contracts and taking a full target exit. It has not been optimized to improve this historical sample. New-policy outcomes must be reported separately from the 37 legacy trades.

Existing limitations remain: decisions settle on completed bars, whole fills are assumed, quote depth and queue execution are not simulated, and stale fallback can produce modeled exits. This is a shadow book, not certified live execution.

Reading the charts

The next pages use stored ES15 sampled-midpoint closes and the 37 recorded fills. IDs identify the same trade across charts and tables. Up triangle = long entry (buy); down triangle = short entry (sell); X = exit (sell long / buy to cover short). Green paths earned money; coral paths lost money.

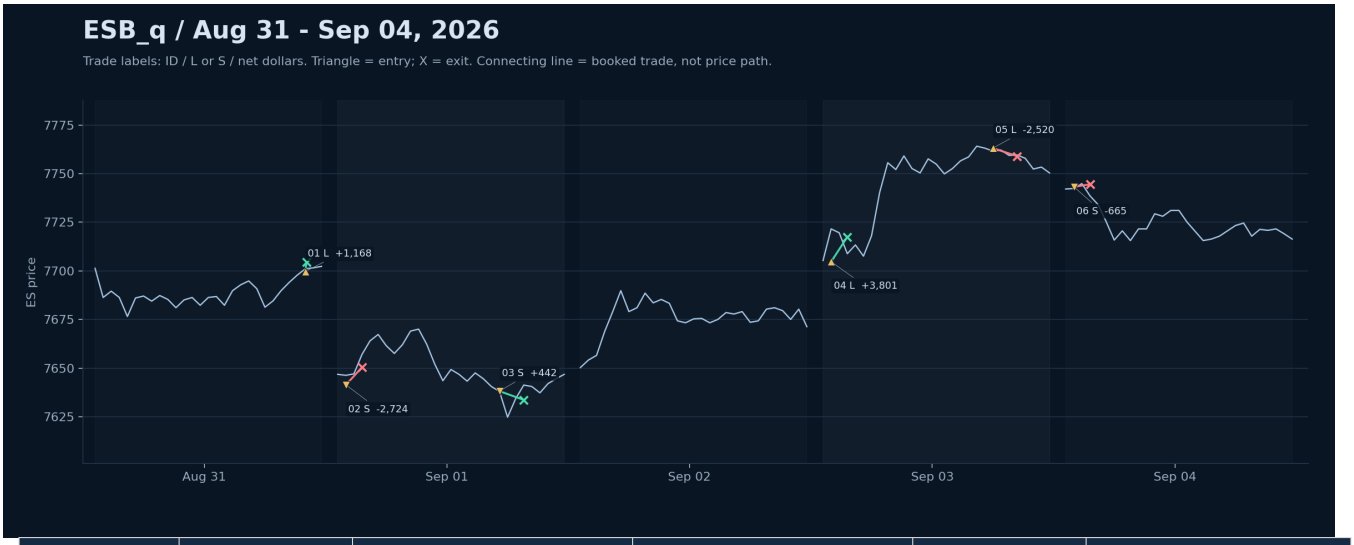
Marker times are ledger processing times, not exact exchange fill timestamps. Missing intervals and overnight/roll gaps are not connected. The full-period view compresses sessions; weekly views and tables make every trade readable.

All recorded trades / full period



Use the separate full-resolution PNG to zoom. Six weekly detail pages follow. All charts show original-policy trades from August 31 through October 9; there are no revised-policy trades in this record.

Aug 31 - Sep 04 / trade detail



ID / day	Side / qty	Entry ET / px	Exit ET / px	Reason	Net
01 08-31	Long / 5	16:00 / 7699.75	16:02 / 7704.50	time	$1,167.50
02 09-01	Short / 6	09:45 / 7641.50	10:15 / 7650.50	ratchet	-$2,724.00
03 09-01	Short / 2	14:30 / 7638.25	15:15 / 7633.75	ratchet	$442.00
04 09-03	Long / 6	09:45 / 7704.75	10:15 / 7717.50	ratchet	$3,801.00
05 09-03	Long / 12	14:45 / 7763.12	15:30 / 7759.00	ratchet	-$2,520.00
06 09-04	Short / 10	09:45 / 7743.25	10:15 / 7744.50	ratchet	-$665.00

Long: buy entry, sell exit. Short: sell entry, buy-to-cover exit. Net includes modeled $4 round-trip commission per contract. No trade prices or outcomes were changed for the chart.

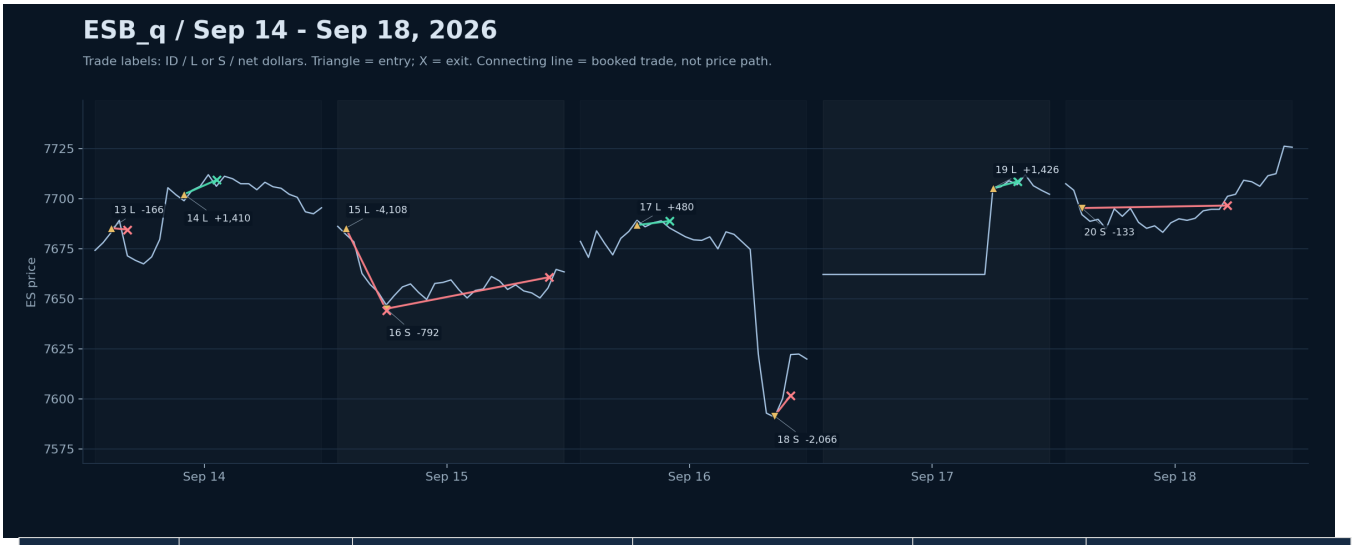
Sep 07 - Sep 11 / trade detail



ID / day	Side / qty	Entry ET / px	Exit ET / px	Reason	Net
07 09-08	Long / 4	09:45 / 7714.00	10:00 / 7693.00	stop	-$4,216.00
08 09-08	Short / 4	11:00 / 7693.38	15:15 / 7688.75	ratchet	$910.00
09 09-08	Short / 7	15:15 / 7688.62	16:02 / 7679.75	time	$3,076.50
10 09-09	Short / 5	11:15 / 7642.75	11:45 / 7635.75	ratchet	$1,730.00
11 09-10	Short / 10	09:45 / 7600.50	10:15 / 7592.00	ratchet	$4,210.00
12 09-11	Long / 5	10:00 / 7680.00	10:30 / 7667.00	ratchet	-$3,270.00

Long: buy entry, sell exit. Short: sell entry, buy-to-cover exit. Net includes modeled $4 round-trip commission per contract. No trade prices or outcomes were changed for the chart.

Sep 14 - Sep 18 / trade detail



ID / day	Side / qty	Entry ET / px	Exit ET / px	Reason	Net
13 09-14	Long / 4	10:00 / 7685.25	10:30 / 7684.50	ratchet	-$166.00
14 09-14	Long / 4	12:15 / 7702.12	13:15 / 7709.25	ratchet	$1,410.00
15 09-15	Long / 2	09:45 / 7685.25	11:00 / 7644.25	stop	-$4,108.00
16 09-15	Short / 1	11:00 / 7645.00	16:02 / 7660.75	time	-$791.50
17 09-16	Long / 5	11:15 / 7686.75	12:15 / 7688.75	ratchet	$480.00
18 09-16	Short / 4	15:30 / 7591.50	16:00 / 7601.75	ratchet	-$2,066.00
19 09-17	Long / 9	14:45 / 7705.25	15:30 / 7708.50	ratchet	$1,426.50
20 09-18	Short / 2	10:00 / 7695.25	14:30 / 7696.50	ratchet	-$133.00

Long: buy entry, sell exit. Short: sell entry, buy-to-cover exit. Net includes modeled $4 round-trip commission per contract. No trade prices or outcomes were changed for the chart.

Sep 21 - Sep 25 / trade detail



ID / day	Side / qty	Entry ET / px	Exit ET / px	Reason	Net
21 09-21	Long / 8	11:00 / 7790.00	11:45 / 7796.25	ratchet	$2,468.00
22 09-23	Short / 10	09:45 / 7824.50	10:15 / 7802.75	ratchet	$10,835.00
23 09-25	Short / 6	09:45 / 7771.00	10:30 / 7761.75	ratchet	$2,751.00
24 09-25	Long / 2	12:15 / 7796.25	16:02 / 7804.75	time	$842.00

Long: buy entry, sell exit. Short: sell entry, buy-to-cover exit. Net includes modeled $4 round-trip commission per contract. No trade prices or outcomes were changed for the chart.

Sep 28 - Oct 02 / trade detail



ID / day	Side / qty	Entry ET / px	Exit ET / px	Reason	Net
25 09-28	Short / 5	10:15 / 7762.25	11:15 / 7733.00	ratchet	$7,292.50
26 09-29	Long / 6	09:45 / 7754.50	10:00 / 7736.00	stop	-$5,574.00
27 09-30	Long / 2	09:45 / 7756.50	10:30 / 7762.75	ratchet	$617.00
28 09-30	Long / 5	11:15 / 7770.75	11:45 / 7768.25	ratchet	-$645.00
29 09-30	Short / 3	14:45 / 7748.75	16:02 / 7715.25	time	$5,013.00
30 10-01	Short / 2	10:15 / 7703.50	10:45 / 7697.75	ratchet	$567.00
31 10-02	Long / 6	09:45 / 7790.00	10:15 / 7785.25	ratchet	-$1,449.00

Long: buy entry, sell exit. Short: sell entry, buy-to-cover exit. Net includes modeled $4 round-trip commission per contract. No trade prices or outcomes were changed for the chart.

Oct 05 - Oct 09 / trade detail



ID / day	Side / qty	Entry ET / px	Exit ET / px	Reason	Net
32 10-05	Long / 9	10:15 / 7800.25	10:45 / 7804.75	ratchet	$1,989.00
33 10-06	Long / 10	09:45 / 7864.50	10:15 / 7878.00	ratchet	$6,710.00
34 10-07	Short / 10	09:45 / 7837.25	10:15 / 7826.75	ratchet	$5,210.00
35 10-08	Short / 2	13:45 / 7803.00	14:15 / 7804.50	ratchet	-$158.00
36 10-09	Short / 7	09:45 / 7838.25	10:15 / 7836.00	ratchet	$759.50
37 10-09	Long / 3	11:30 / 7853.75	13:30 / 7853.50	ratchet	-$49.50

Long: buy entry, sell exit. Short: sell entry, buy-to-cover exit. Net includes modeled $4 round-trip commission per contract. No trade prices or outcomes were changed for the chart.